

High-Speed 8-bit Time-Interleaved Asynchronous SAR ADC in 180 nm CMOS Technology

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Abstract—This paper presents the design and simulation of a high-speed 8-bit SAR ADC targeting 250 MHz signal bandwidth in IBM 180 nm CMOS. To meet $f_s = 500$ MS/s without requiring a single-slice clock above 4 GHz, an 8-way time-interleaved (TI) architecture is adopted, each slice running an asynchronous SAR at 62.5 MHz. Sub-blocks—bootstrapped S/H, differential capacitive DAC, StrongARM comparator, and Verilog-A SAR/TDM logic—are verified individually then integrated. Post-integration results: ENOB = 6.94 bits, SNR = 43.55 dB, FOM = 148.49 dB at 8 mW.

Index Terms—SAR ADC, time-interleaved, asynchronous SAR, bootstrapped switch, StrongARM comparator, capacitive DAC, 180 nm CMOS.

I. INTRODUCTION

SAR ADCs are favored for energy efficiency and digital scalability. Extending bandwidth beyond 100 MHz at 8-bit resolution introduces two bottlenecks: the binary-search algorithm demands $> 8 \times f_s$ internal clocking in a synchronous topology, and the capacitive DAC must settle within a tight time budget. Running a single SAR above 4 GHz is impractical in 180 nm. Two techniques address this: (1) an *asynchronous* SAR that self-times each bit decision within one f_s cycle, and (2) *time-interleaving* eight slices so each runs at only $f_s/8 = 62.5$ MHz. This work presents the full design flow from transistor-level sub-block characterization to integrated TI-SAR simulation.

II. ARCHITECTURE OVERVIEW

Fig. 1 shows the top-level TI-SAR ADC. Eight identical SAR slices share the differential input and are sequenced by a Verilog-A TDM at $f_{clk} = 500$ MHz. Each slice receives a phase-shifted enable ($\phi_0 \dots \phi_7$) and completes all eight SAR iterations within its 16 ns window. The TDM output drives an ideal reconstruction DAC for FFT evaluation.

III. SUB-BLOCK DESIGN AND CHARACTERIZATION

A. Bootstrapped Sample-and-Hold

The bootstrapped switch maintains constant R_{ON} by bootstrapping the gate to $V_{in} + V_{DD}$, ensuring linearity over the full 1.2 V swing at $V_{CM} = 900$ mV. The S/H drives the differential cap-DAC array ($C_{total} = 256 \times 5$ fF = 1.28 pF plus TG parasitics).

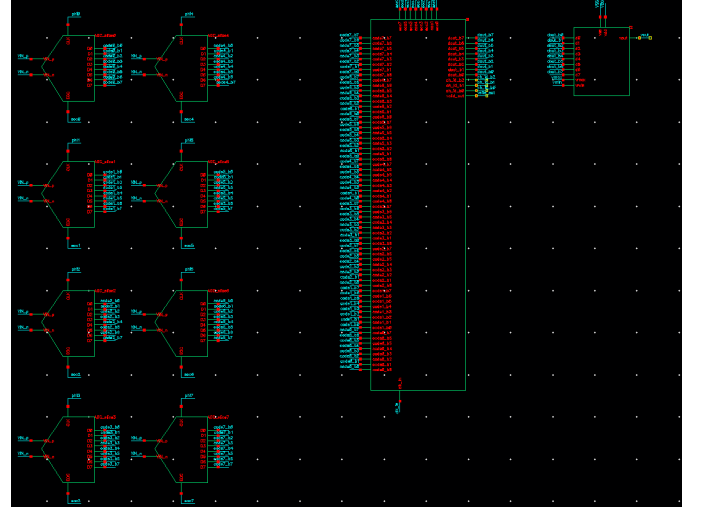
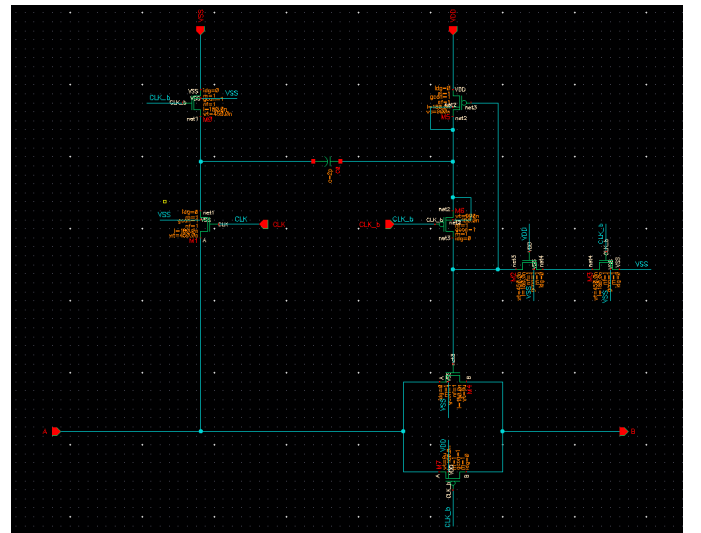
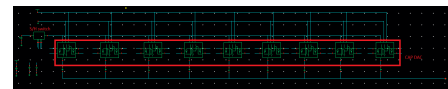


Fig. 1. TI-SAR ADC: 8 slices (ϕ_0 – ϕ_7), Verilog-A SAR/TDM, and ideal output DAC.



(a) S/H + cap-DAC schematic



(b) Characterization testbench

Fig. 2. S/H circuit and testbench (DAC code 11111111, $C = 1.28$ pF; $f_s = 250$ MHz, $f_{in} = 31.25$ MHz).

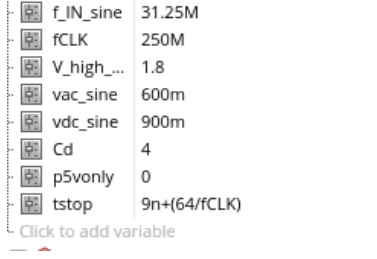


Fig. 3. S/H testbench variables: $f_{CLK} = 250$ MHz, $f_{IN} = 31.25$ MHz, $V_{CM} = 900$ mV, $V_{AC} = 600$ mV, $C_d = 5$ fF.

Waveforms in Fig. 4 confirm tracking during clock-high and holding during clock-low; the bootstrapped gate consistently exceeds V_{DD} . A 64-point FFT (Fig. 5) gives $ENOB = 9.24$ bits, $SNR = 57.43$ dB—over 1 bit of headroom above target.

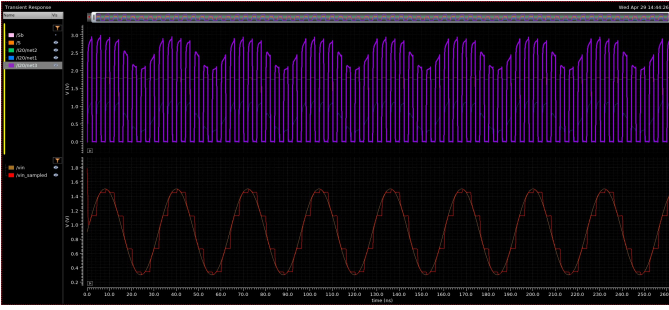


Fig. 4. S/H waveforms: bootstrapped gate (violet, $> V_{DD}$), sampled output tracking/holding input correctly.

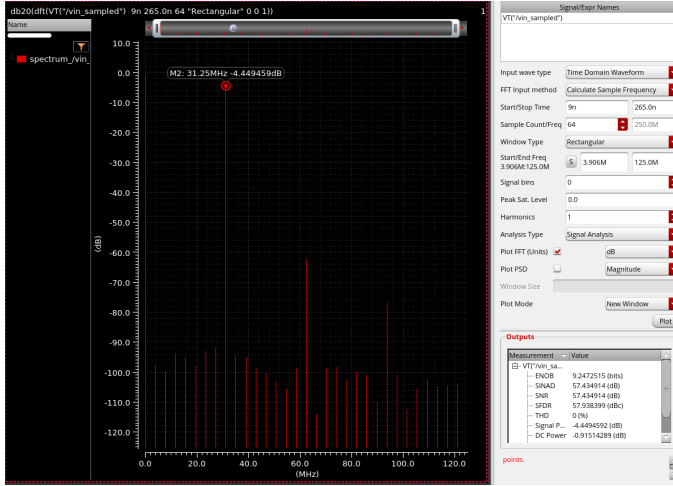
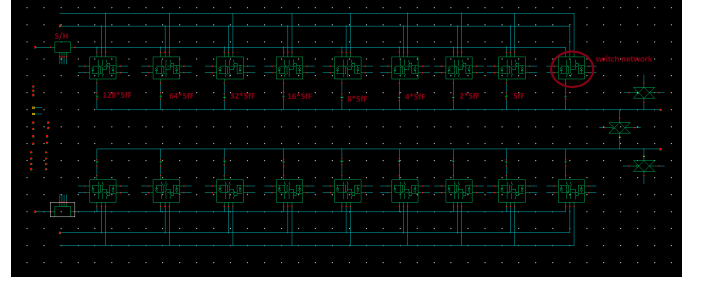


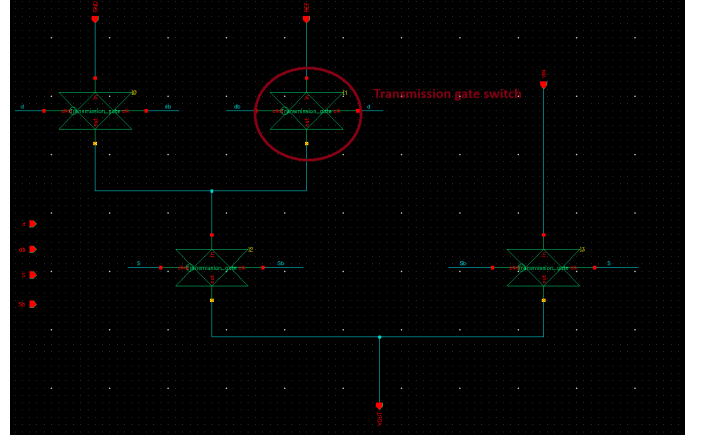
Fig. 5. 64-pt FFT of S/H output at 31.25 MHz: $ENOB = 9.24$ bits, $SNR = 57.43$ dB.

B. Differential Capacitive DAC

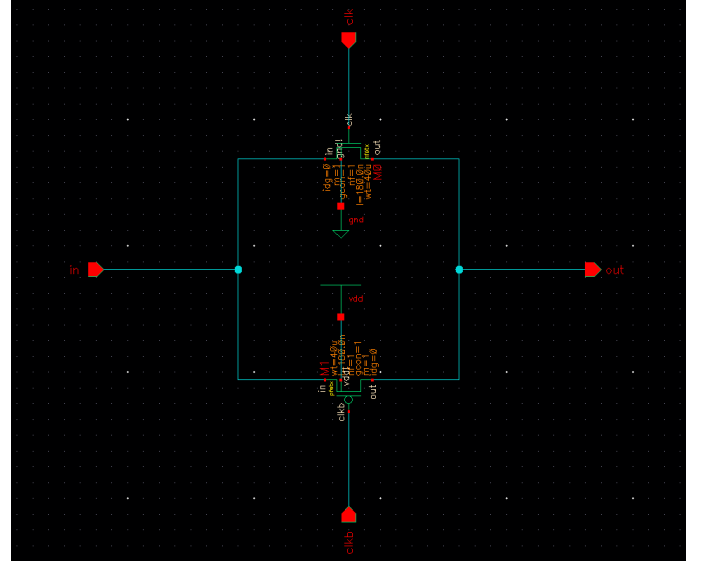
The binary-weighted array (Fig. 6) uses $C_u = 5$ fF unit caps with transmission-gate switches (Figs. 6b, 6c). References $V_{low} = 300$ mV, $V_{high} = 1.5$ V give $V_{FS} = 1.2$ V and $1 \text{ LSB} \approx 4.7$ mV.



(a) Differential cap-DAC schematic



(b) Switch network



(c) Transmission gate

Fig. 6. Cap-DAC: (a) full binary-weighted differential array, (b) switch network, (c) transmission gate cell.

Fig. 7 verifies S/H settling to 900 mV within 2 ns. DAC code sweep (Fig. 8) and settling transients (Fig. 9) confirm every transition reaches within 0.5 LSB (< 2.3 mV) in < 450 ps, meeting the 600 ps comparison budget. Table I lists measured vs. ideal outputs.



Fig. 7. S/H settling to 900 mV within 2 ns ($V_{in} = 900$ mV).

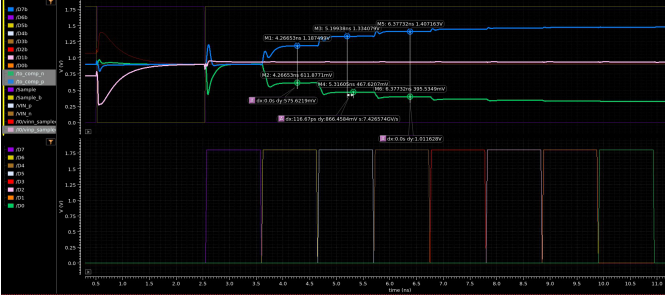


Fig. 8. DAC output during MSB→LSB code sweep ($V_{in,p} = V_{in,n} = V_{CM}$).

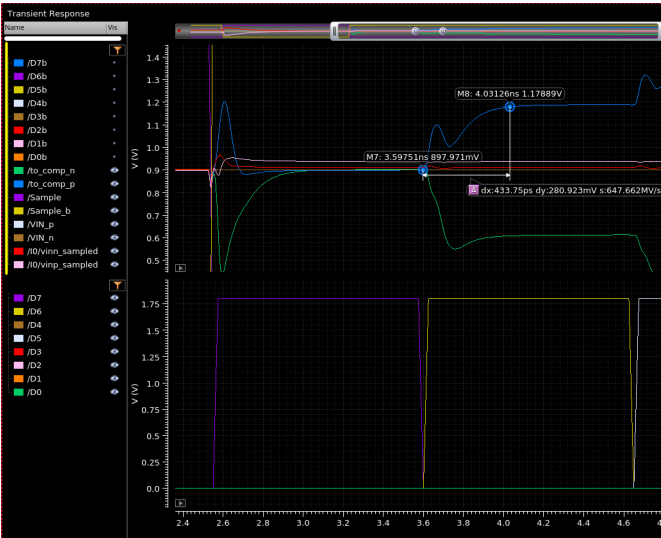


Fig. 9. DAC settling per bit decision: all within 0.5 LSB in < 450 ps.

TABLE I
CAP-DAC OUTPUT VS. IDEAL ($V_{in} = V_{CM}$)

DAC Code	Measured	Ideal
10000000	0 mV	0 mV
01000000	575.62 mV	594.36 mV
00100000	866.45 mV	893.88 mV
00010000	1010 mV	1043 mV

C. StrongARM Comparator with Pre-Amplifier

Fig. 10 shows the comparator: a $\approx \times 4$ pre-amplifier (red box) followed by a StrongARM latch and RS de-glitch latch. The pre-amp raises 100 mV differential to ≈ 450 mV, reducing kickback and metastability risk.

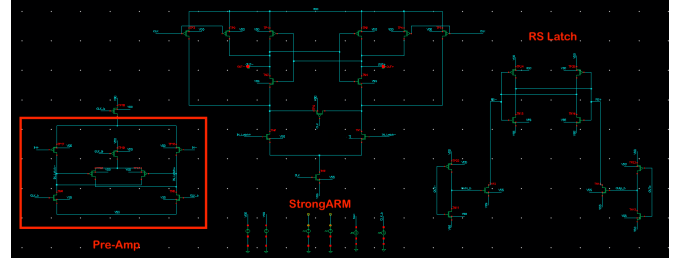


Fig. 10. Comparator: pre-amp (red, $\approx \times 4$), StrongARM latch, RS output latch.

Timing waveforms (Fig. 11) at 1 GHz show clock-to-comparator delay 230 ps and clock-to-latch 380 ps. In the integrated ADC the comparator runs at 830 MHz (600 ps window).

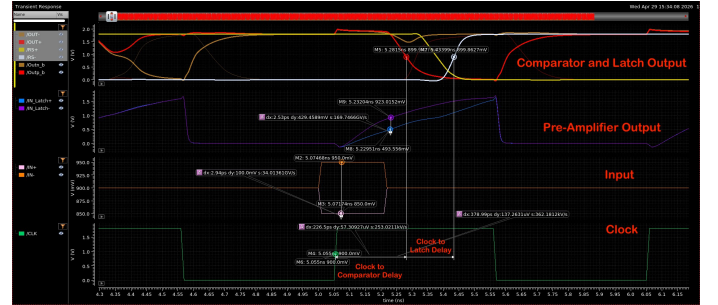


Fig. 11. Comparator timing (100 mV input, 1 GHz clock): pre-amp $\approx \times 4$; clk-to-out 230 ps; clk-to-latch 380 ps.

Systematic offset is found by ramping $V_{IN\pm}$ to the metastability crossing (Fig. 12): $V_{os,sys} = 2.11$ mV < 0.5 LSB. Monte Carlo over 100 samples (Fig. 13) gives $\sigma = 3.28$ mV, $\mu = 2.28$ mV.

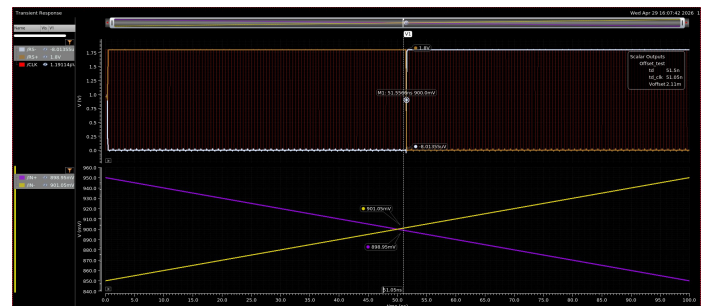


Fig. 12. Systematic offset via ramp test: $V_{os,sys} = 2.11$ mV.

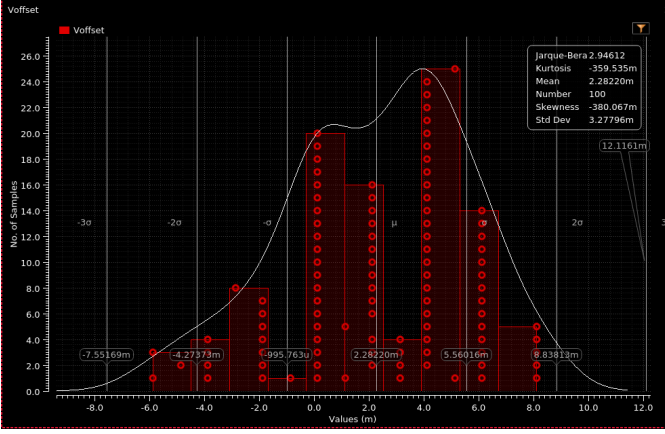


Fig. 13. Monte Carlo offset (100 runs): $\sigma = 3.28$ mV, $\mu = 2.28$ mV.

IV. SINGLE-SLICE ADC SIMULATION

The single-slice integrates all blocks above with Verilog-A asynchronous SAR logic; a re-timer emulates TDM output. Fig. 14 shows the full schematic. Simulation variables are in Fig. 15.

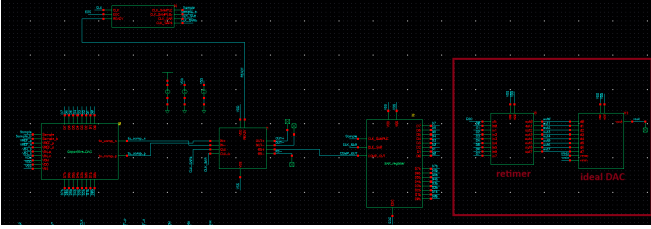


Fig. 14. Single-slice SAR ADC schematic: bootstrapped S/H, cap-DAC, StrongARM comparator, and Verilog-A asynchronous SAR logic with re-timer.

	Name	Value
1	Cd	5
2	fCLK	62.5M
3	V_high_CLK	1.8
4	vac_sine	600m
5	vdc_sine	900m
6	p5only	0
7	k	29
8	tstop	150n+(256/fCLK)
9	f_IN_sine	(k/256)*fCLK

Fig. 15. Single-slice variables: $f_{CLK} = 62.5$ MHz, $C_d = 5$ fF, $f_{in} = (k/256) \cdot f_{CLK}$.

Fig. 16 shows $V_{IN,P}$ and the sampled node. Fig. 17 captures one 16 ns conversion resolving 01100001; Fig. 18 details the 8-cycle asynchronous SAR clock. A 128-point FFT (Fig. 19) gives ENOB = 6.91 bits, SNR = 43.37 dB. Per-slice power: $604.7 \mu A \times 1.8 V \approx 1$ mW; total for eight slices ≈ 8 mW.

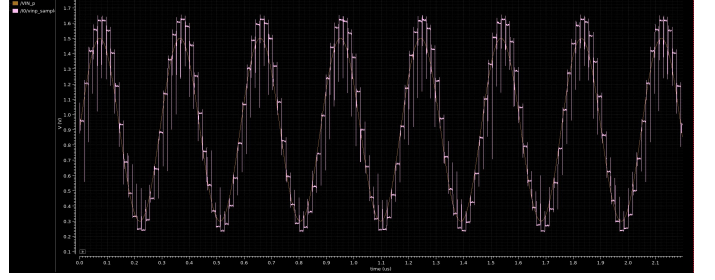


Fig. 16. $V_{IN,P}$ (orange) and sampled node (pink) at 62.5 MHz showing per-cycle hold steps.

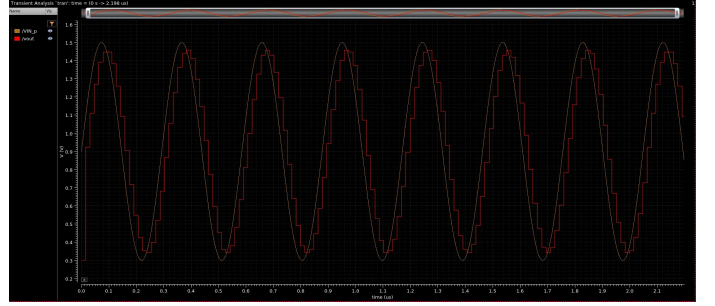


Fig. 17. single slice with clock = 62.5 MHz and input sine of $7/128 \cdot 62.5$ MHz = 3.417 MHz

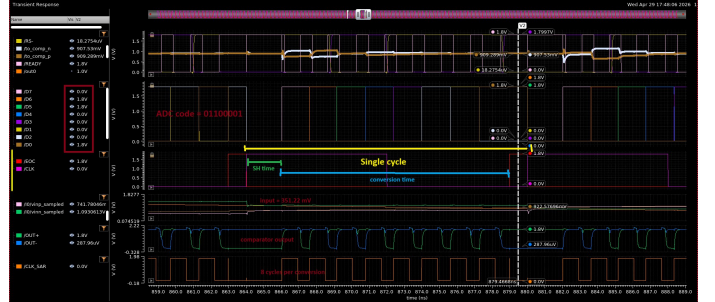


Fig. 18. Single-slice waveforms over 16ns, ADC code 01100001: comparator toggles and 8-cycle asynchronous SAR clock per conversion window.

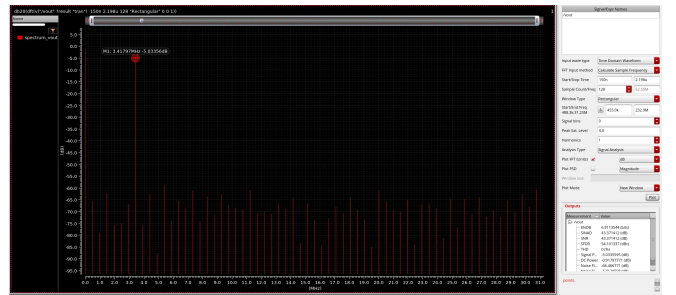


Fig. 19. 128-pt FFT, single slice ($f_{in} = 3.42$ MHz, $f_{clk} = 62.5$ MHz): ENOB = 6.91 bits, SNR = 43.37 dB.

V. INTEGRATED TI-SAR ADC RESULTS

Eight slices (Fig. 20) are assembled with the Verilog-A TDM at $f_{clk} = 500$ MHz. Input and reconstructed output at 13.6 MHz are shown in Fig. 21; the 256-point FFT (Fig. 22) yields ENOB = 6.94 bits, SNR = 43.55 dB. Full-bandwidth operation at 250 MHz Nyquist is confirmed in Fig. 23.

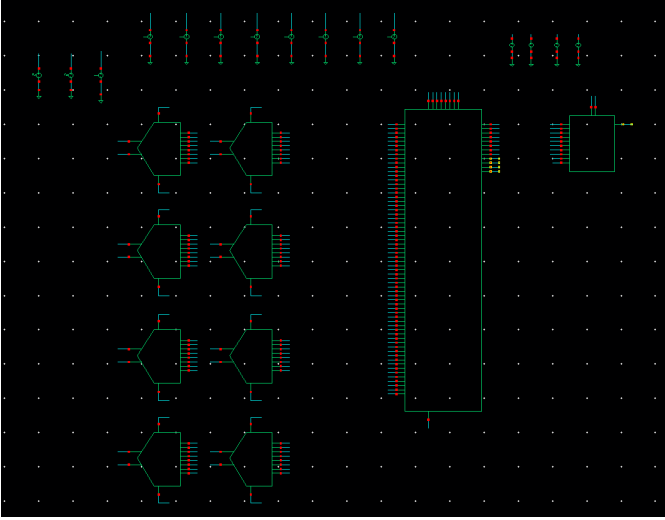


Fig. 20. Integrated TI-SAR feeding TDM output logic.

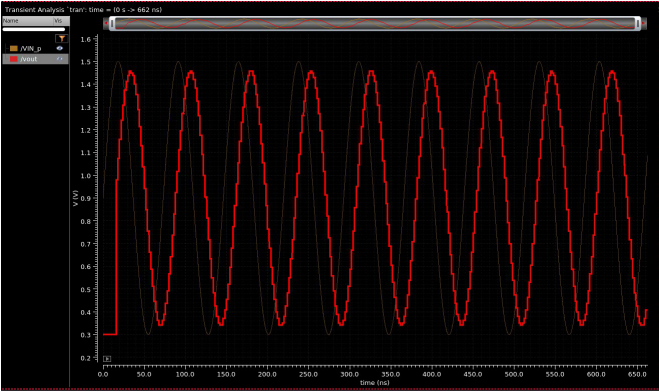


Fig. 21. TI-SAR transient: V_{IN} (gold) and reconstructed output (red), $f_{in} = 13.6$ MHz, $f_{clk} = 500$ MHz.

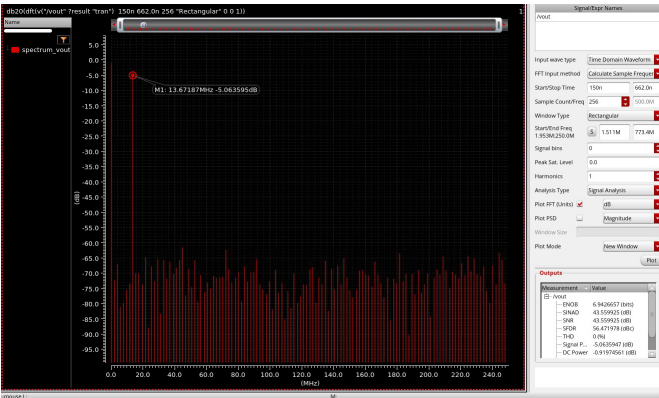


Fig. 22. 256-pt FFT: ENOB = 6.94 bits, SNR = 43.55 dB, SFDR = 56.47 dBc.

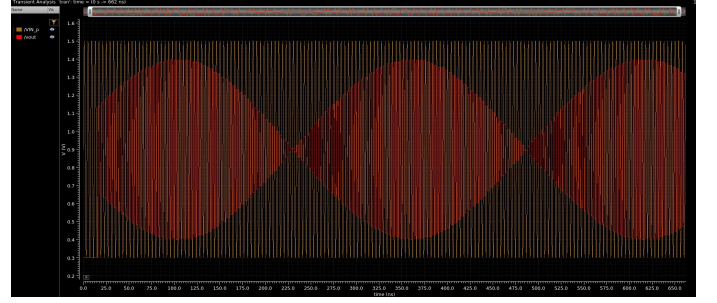


Fig. 23. Nyquist test: $f_{in} = 250$ MHz, $f_{clk} = 500$ MHz, full-BW operation confirmed.

A. Figure of Merit

The Schreier FOM is defined as:

$$\text{FoM}_S = \text{SNDR} + 10 \log_{10} \frac{\text{BW}}{P} = 148.49 \text{ dB.} \quad (1)$$

Table II compares achieved vs. target specs.

TABLE II
PERFORMANCE SUMMARY

Parameter	Target	Achieved
Signal BW	250 MHz	250 MHz
Sampling Rate	500 MS/s	500 MS/s
Resolution	8 bit	8 bit
ENOB	8 bits	6.94 bits
SNR	—	43.55 dB
SFDR	—	56.47 dBc
Power	—	≈ 8 mW
FoM _S	> 170 dB	148.49 dB
Technology	180 nm	180 nm

VI. CONCLUSION

An 8-way TI asynchronous SAR ADC was designed and verified in 180 nm CMOS. Eight slices at 62.5 MHz deliver 500 MS/s over 250 MHz bandwidth. Post-integration FFT gives ENOB = 6.94 bits, SNR = 43.55 dB at ≈ 8 mW ($\text{FoM}_S = 148.49$ dB). Sub-blocks individually meet targets (S/H: 9.24 ENOB; DAC: < 450 ps settling; comparator: $V_{os} = 2.11$ mV), pointing to inter-slice calibration as the primary path to the 8-ENOB, 170 dB FOM goal.

REFERENCES

- [1] M. Dessouky and A. Kaiser, "Very low-voltage digital-audio delta-sigma modulator with 88-dB dynamic range using local switch bootstrapping," *IEEE J. Solid-State Circuits*, vol. 36, no. 3, pp. 349–355, Mar. 2001.
- [2] K. Bandla and D. Pal, "Strong-ARM dynamic latch comparators: Design and analyses on CAD platform," Dept. EEE, BITS-Pilani.
- [3] N. Sharma, V. Hande, and D. M. Das, "A dynamic comparator with cross-coupled pre-amplifier with <160 ps delay and 81 fJ/ns EDP," *IEEE Trans. Circuits Syst. I*, 2021.
- [4] W. Mao, Y. Li, C. H. Heng, and Y. Lian, "A low-power 12-bit 1-kS/s SAR ADC for biomedical signal processing," *IEEE Trans. Circuits Syst. I*, vol. 66, no. 2, pp. 477–488, Feb. 2019.
- [5] B. Razavi, "The StrongARM latch," *IEEE Solid-State Circuits Mag.*, vol. 7, no. 2, pp. 12–17, Spring 2015.
- [6] O. Kardoni, "A study of SAR ADC and implementation of 10-bit asynchronous design," M.S. thesis, 2013.
- [7] P. J. A. Harpe *et al.*, "A 26 μ W 8-bit 10 MS/s asynchronous SAR ADC for low-energy radios," *IEEE J. Solid-State Circuits*, vol. 47, no. 7, pp. 1585–1595, Jul. 2012.
- [8] Z. Liu, B. Nikolic, and V. Stojanovic, "Time-interleaved SAR ADC design using Berkeley Analog Generator," in *Proc. IEEE CICC*, 2020.